SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	V.Aadhithya	**Reg. No.:**	192511256
Section / Batch:	B.E. -CSE	**Date:**	20/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:		**100 Marks**

SECTION A – CONCEPT MAPPING

Q3, Consider the concept map: Recurrence Relation → Substitution Method → Complexity Derivation Trace the path from a given recurrence to its final closed-form complexity. Add intermediate nodes such as Guess → Induction → Verification, and explain how each connection in the extended map contributes to determining algorithm efficiency.

Q43) Consider the concept map: Algorithm → Space complexity → Auxiliary space → Input space Map how memory usage is calculated. Extend the map with time-space trade-off. Justify when higher space is acceptable. Interpret how the map helps design efficient solutions.

Code of Conduct

I V.Aadhithya (192511256) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: V.Aadhithya

RUBRICS FOR CALCULATION

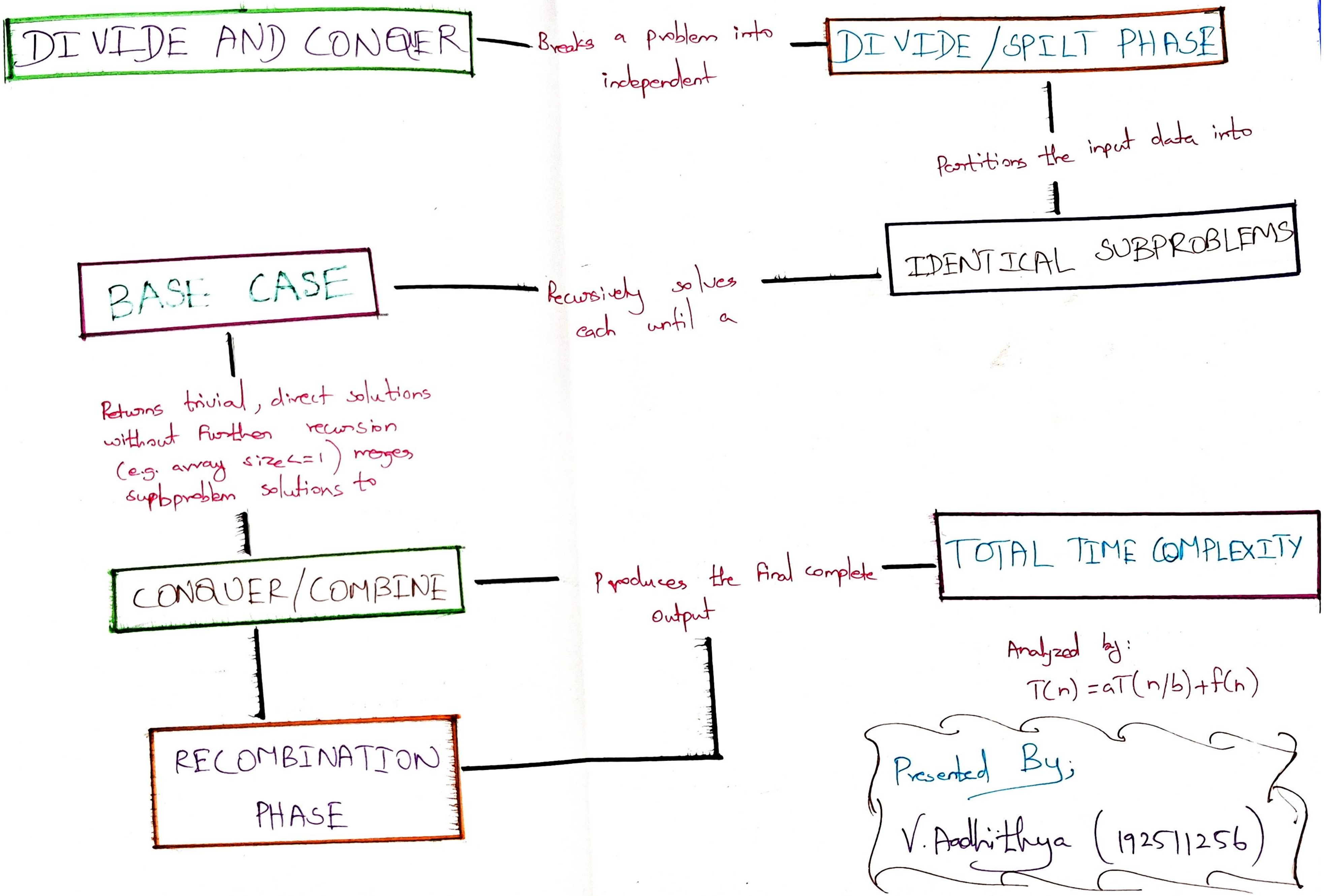
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

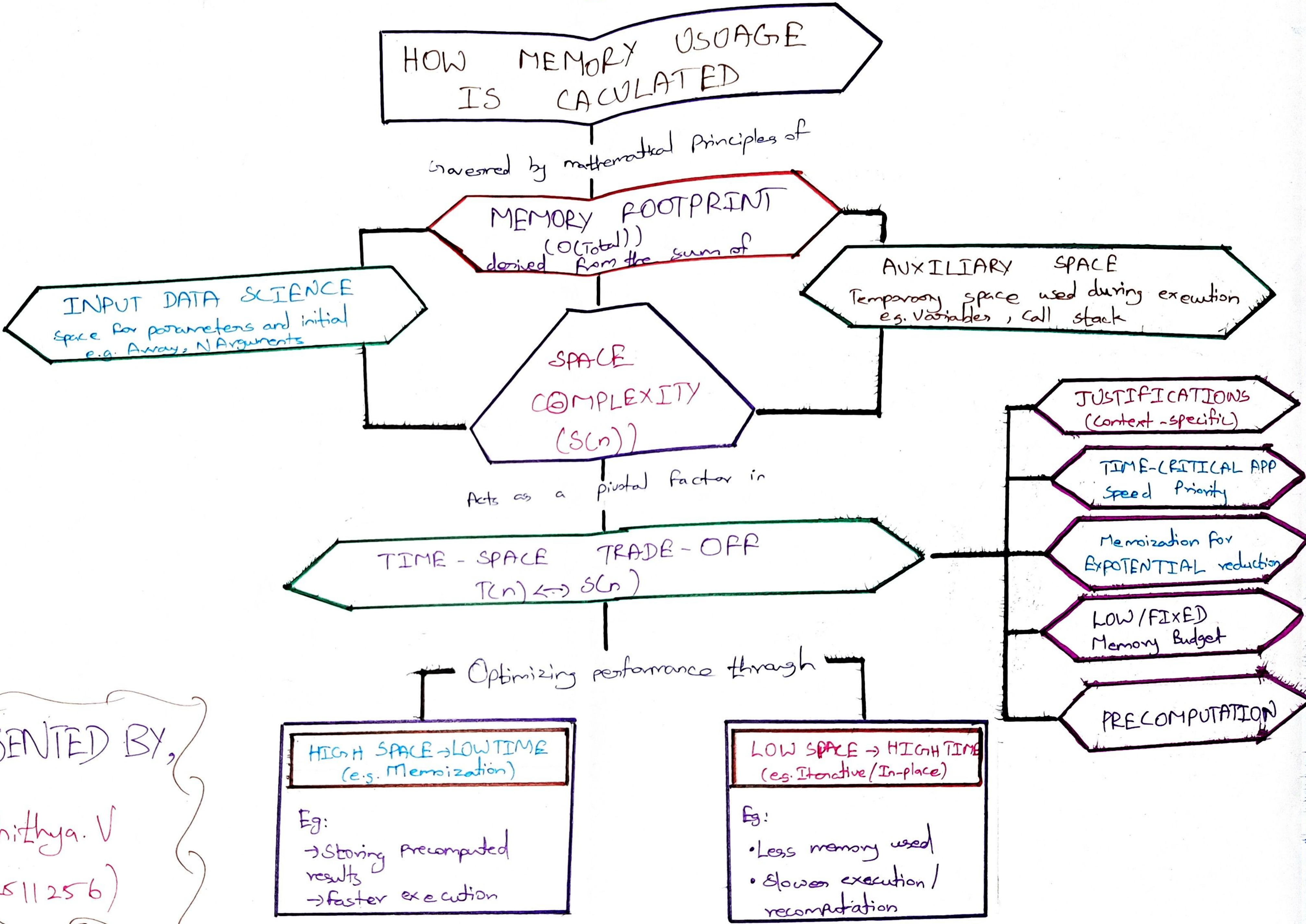
Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1	3	3	3	3	4	39
Q2	3	3	3	3	4	39
					Overall Total (100)	78

Faculty In-charge	Course Coordinator	Head of Department
Signature: [signature]	Signature: ______________	Signature: ______________
Date: 20/7/2026	Date: ______________	Date: ______________

CONCEPT MAP


DIVIDE AND CONQUER
Breaks a problem into independent
DIVIDE / SPILT PHASE
Partitions the input data into
IDENTICAL SUBPROBLEMS
Recursively solves each until a
BASE CASE
Returns trivial, direct solutions without further recursion (e.g. array size<=1) merges supbproblem solutions to
CONQUER/COMBINE
RECOMBINATION PHASE
Produces the final complete output
TOTAL TIME COMPLEXITY
Analyzed by:
T(n) =aT(n/b)+f(n)
Presented By;
V. Aadhithya (192511256)

CONCEPT MAP



PRESENTED BY,

Aadhithya. V

(192511256)